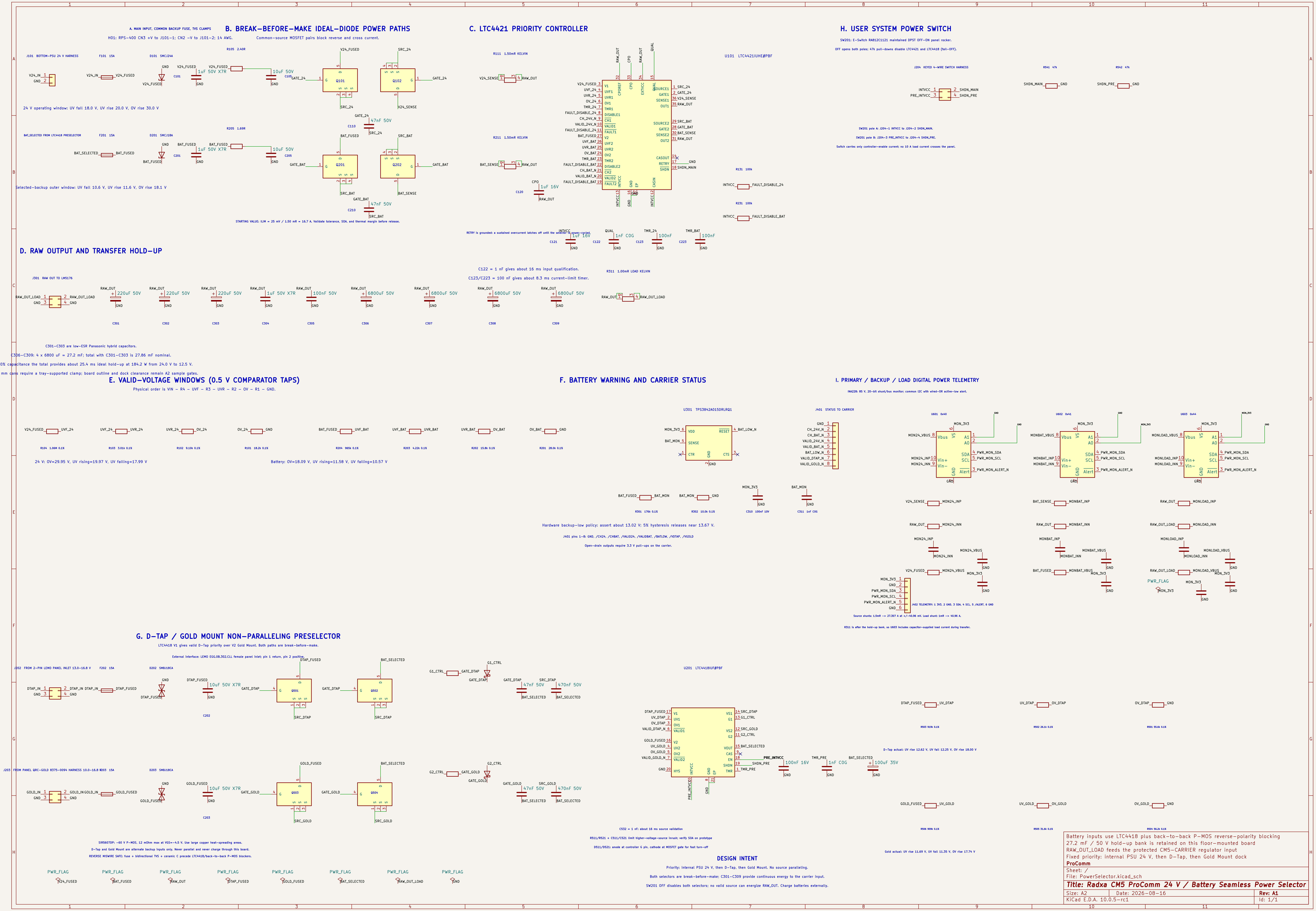

A. MAIN INPUT, COMMON BACKUP FUSE, TVS CLAMPS

H01: RPS-400 CN3 +V to J101-1; CN2 -V to J101-2; 14 AWG.

24 V operating window: UV fall 18.0 V, UV rise 20.0 V, OV rise 30.0 V

Selected-backup outer window: UV fall 10.6 V, UV rise 11.6 V, OV rise 18.1 V

B. BREAK–BEFORE–MAKE IDEAL–DIODE POWER PATHS

Common-source MOSFET pairs block reverse and cross current.

STARTING VALUE: ILIM = 25 mV / 1.50 mR = 16.7 A. Validate tolerance, SOA, and thermal margin before release.

C. LTC4421 PRIORITY CONTROLLER

RETRY is grounded: a sustained overcurrent latches off until the selector is power-cycled.

C122 = 1 nF gives about 16 ms input qualification.

C123/C223 = 100 nF gives about 8.3 ms current-limit timer.

H. USER SYSTEM POWER SWITCH

SW201: E-Switch RA812C1121 maintained DPST OFF-ON panel rocker.

OFF opens both poles; 47k pull-downs disable LTC4421 and LTC4418 (fail-OFF).

SW201 pole A: J204-1 INTVCC to J204-2 SHDN_MAIN.

SW201 pole B: J204-3 PRE_INTVCC to J204-4 SHDN_PRE.

Switch carries only controller-enable current; no 10 A load current crosses the panel.

D. RAW OUTPUT AND TRANSFER HOLD–UP

C301–C303 are low-ESR Panasonic hybrid capacitors.

C306–C309: 4 x 6800 uF = 27.2 mF; total with C301–C303 is 27.86 mF nominal.

At -20% capacitance the total provides about 25.4 ms ideal hold-up at 184.2 W from 24.0 V to 12.5 V.

x 35 mm cans require a tray-supported clamp; board outline and dock clearance remain A2 sample gates.

E. VALID–VOLTAGE WINDOWS (0.5 V COMPARATOR TAPS)

Physical order is VIN – R4 – UVF – R3 – UVR – R2 – OV – R1 – GND.

24 V: OV=29.95 V, UV rising=19.97 V, UV falling=17.99 V

Battery: OV=18.09 V, UV rising=11.58 V, UV falling=10.57 V

F. BATTERY WARNING AND CARRIER STATUS

Hardware backup-low policy: assert about 13.02 V; 5% hysteresis releases near 13.67 V.

J401 pins 1–8: GND, /CH24, /CHBAT, /VALID24, /VALIDBAT, /BATLOW, /VDTAP, /VGOLD

Open-drain outputs require 3.3 V pull-ups on the carrier.

I. PRIMARY / BACKUP / LOAD DIGITAL POWER TELEMETRY

INA228: 85 V, 20-bit shunt/bus monitor; common I2C with wired-OR active-low alert.

J402 TELEMETRY: 1 3V3, 2 GND, 3 SDA, 4 SCL, 5 /ALERT, 6 GND

Source shunts: 1.5mR -> 27.307 A at +/-40.96 mV. Load shunt: 1mR -> 40.96 A.

R311 is after the hold-up bank, so U603 includes capacitor-supplied load current during transfer.

G. D–TAP / GOLD MOUNT NON–PARALLELING PRESELECTOR

LTC4418 V1 gives valid D-Tap priority over V2 Gold Mount. Both paths are break-before-make.

External interface: LEMO EGG.0B.302.CLL female panel inlet; pin 1 return, pin 2 positive.

D-Tap actual: UV rise 12.62 V, UV fall 12.25 V, OV rise 18.00 V

Gold actual: UV rise 11.69 V, UV fall 11.35 V, OV rise 17.74 V

SIR5607DP: -60 V P-MOS, 12 mOhm max at VGS=-4.5 V. Use large copper heat-spreading areas.

D-Tap and Gold Mount are alternate backup inputs only. Never parallel and never charge through this board.

REVERSE MISWIRE SAFE: fuse + bidirectional TVS + ceramic C precede LTC4418/back-to-back P-MOS blockers.

C532 = 1 nF: about 16 ms source validation

R511/R521 + C511/C521 limit higher-voltage-source inrush; verify SOA on prototype

D511/D521: anode at controller G pin, cathode at MOSFET gate for fast turn-off

DESIGN INTENT

Priority: internal PSU 24 V, then D-Tap, then Gold Mount. No source paralleling.

Both selectors are break-before-make; C301–C309 provide continuous energy to the carrier input.

SW201 OFF disables both selectors; no valid source can energize RAW_OUT. Charge batteries externally.

Battery inputs use LTC4418 plus back-to-back P-MOS reverse-polarity blocking
27.2 mF / 50 V hold-up bank is retained on this floor-mounted board
RAW_OUT_LOAD feeds the protected CM5-CARRIER regulator input
Fixed priority: internal PSU 24 V, then D-Tap, then Gold Mount dock
ProComm
Sheet: /
File: PowerSelector.kicad_sch
Title: Radxa CM5 ProComm 24 V / Battery Seamless Power Selector
Size: A2 Date: 2026-08-16 **Rev: A1**
KiCad E.D.A. 10.0.5-rc1 Id: 1/1